

SCHÄFFER'S CONJECTURE FOR $k = 14$

IMIN CHEN, STEPHEN CHOI, AND JAMES HOULE

ABSTRACT. We extend the results of Bennett-Györy-Pintér to prove Schäffer's conjecture for $k = 14$.

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1. INTRODUCTION

Schäffer's conjecture [Sch56] concerns the following diophantine problem:

Conjecture 1.1. Let $S_k(x) = 1^k + 2^k + \cdots + x^k$. Then the only solutions to the equation

$$(1) \quad S_k(x) = y^n$$

in $x, y, k, n \in \mathbb{N}$ and $n \geq 2$ is the trivial solution $(x, y, k, n) = (1, 1, k, n)$ unless

$$(2) \quad (k, n) \in \{(1, 2), (3, 2), (3, 4), (5, 2)\},$$

in which case there are infinitely many solutions in $x, y \in \mathbb{N}$, or

$$(3) \quad (k, n, x, y) = (2, 2, 24, 70).$$

By a descent argument, it is shown in [Sch56, Theorem 1] that (1) reduces to the integer solutions on a curve of positive genus when (k, n) is not in the list of (2), from which Siegel's Theorem shows there are only finite many solutions in $x, y \in \mathbb{N}$. This is a weaker statement than Schäffer's conjecture but nonetheless gives an explanation for the exceptional cases in (2).

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For each exceptional case (k, n) in (2), it is shown in [Sch56, Theorem 1], again by descent methods, that there are infinitely many solutions in $x, y \in \mathbb{N}$.

The most comprehensive result on Schäffer's conjecture thus far is the following theorem, which uses most of the diophantine methods in the modern toolkit:

Theorem 1.2. (*Bennett-Györy-Pintér* [BGP04]) For $1 \leq k \leq 11$, equation (1) has only the trivial solution, unless $(k, n) \in \{(1, 2), (3, 2), (3, 4), (5, 2)\}$, or $k = 2$, in which case there is the additional solution $(n, x, y) = (2, 24, 70)$.

We prove in this paper a new case of Schäffer's conjecture with the following theorem.

Theorem 1.3. When $k = 14$, there are no solutions to (1) in $x, y, n \in \mathbb{N}$ and $n \geq 2$.

A new ingredient in our proof is that we apply descent using all three factors $x, x + 1, 2x + 1$ present in the formula for $S_k(x)$ and we systematically apply the modular method first, including the use of the multi-Frey technique, Kraus' method, local methods modulo a power of a prime, and a result of Darmon-Merel [DM97]. A small number of cases then remain, but these are dealt with by prior results on ternary equations with coefficients [Ben01] [BBGP10]. Another result we use is [JPW03] which verifies Schäffer's conjecture when $k \leq 58$ is even and $n = 2$.

Remark if $n > 1$, then n is divisible by a prime p , and thus to study (1) it suffices to consider $n = p$ prime. Henceforth, we will mainly deal with prime exponents $n = p$.

2. DESCENT STEP

For even k , we may write

$$(4) \quad S_k(x) = \frac{1}{C_k} x(x+1)(2x+1)T_k(x)$$

where $C_k \in \mathbb{Z}$. Let $D = \gcd(x(x+1), T_k(x))$ and $d = \gcd(2x+1, T_k(x))$ where we consider the gcd over $\mathbb{Z}$ for a particular $x \in \mathbb{Z}$. For a fixed k , we will use the following result to get a finite list of possible values for D and d .

Lemma 2.1. Let $f = \alpha x + \beta, g = \gamma x^e + \dots \in \mathbb{Z}[x]$, $\alpha, \gamma, \delta \neq 0$ be such that the absolute value of the resultant $|\text{Res}(f, g, x)| = \delta$ where δ is squarefree. Then for all $x_0 \in \mathbb{Z}$, $\gcd(f(x_0), g(x_0)) \mid \delta$. Furthermore, suppose $\gcd(\alpha, \delta) = 1$ and $\delta \mid \alpha^{e-1} g'(-\frac{\beta}{\alpha})$. Then if a prime $p \mid \gcd(f(x_0), g(x_0))$, then $p \parallel g(x_0)$.

Proof. Recall that there are polynomials $A(x)$ and $B(x)$ in $\mathbb{Z}[x]$ such that

$$A(x)f(x) + B(x)g(x) = \text{Res}(f, g, x) = \alpha^e g\left(-\frac{\beta}{\alpha}\right).$$

Suppose $x_0 \in \mathbb{Z}$. We have that $\gcd(f(x_0), g(x_0)) \mid \delta$.

Suppose a prime $p \mid \gcd(f(x_0), g(x_0))$. Consider $\alpha^e g\left(\frac{x-\beta}{\alpha}\right)$ as a polynomial of degree e in $\mathbb{Z}[x]$ and write

$$\alpha^e g\left(\frac{x-\beta}{\alpha}\right) = \alpha^e g\left(-\frac{\beta}{\alpha}\right) + \alpha^{e-1} g'\left(-\frac{\beta}{\alpha}\right)x + x^2 p(x)$$

for some $p(x) \in \mathbb{Z}[x]$. Then for any $x \in \mathbb{Z}$, we have

$$\alpha^e g\left(\frac{px - \beta}{\alpha}\right) \equiv \alpha^e g\left(-\frac{\beta}{\alpha}\right) + \alpha^{e-1} g'\left(-\frac{\beta}{\alpha}\right)(px) \equiv \alpha^e g\left(-\frac{\beta}{\alpha}\right) = \text{Res}(f, g, x) \pmod{p^2}$$

because $p \mid \delta$ and $\delta \mid \alpha^{e-1} g'\left(-\frac{\beta}{\alpha}\right)$. Therefore, $p^2 \nmid \alpha^e g\left(\frac{px - \beta}{\alpha}\right)$ because $\text{Res}(f, g, x)$ is square-free. We have proved that $p \parallel \alpha^e g\left(\frac{px - \beta}{\alpha}\right)$ for all $x \in \mathbb{Z}$. Since $p \mid f(x_0) = \alpha x_0 + \beta$, we write $x_0 = \frac{px - \beta}{\alpha}$ for some $x \in \mathbb{Z}$. So $p \parallel g(x_0)$ because $\gcd(\alpha, \delta) = 1$. This proves Lemma 2.1. $\square$

Corollary 2.2. For $2 \leq k \leq 14$ even, we have that

- (i) D and d are both square-free,
- (ii) if a prime $p \mid D$ or $p \mid d$, then $p \parallel T_k(x)$.

Proof. We apply Lemma 2.1 to each linear factor, $x, x+1, 2x+1$, where we note the three linear factors are pairwise coprime. We need to check the condition that $|\text{Res}(f, g, x)| = \delta \mid \alpha^{\deg(T_k)} T'_k\left(-\frac{\beta}{\alpha}\right)$. For example, for $k = 14$, we have

$$T_{14}(x) = 3x^{12} + 18x^{11} + 24x^{10} - 45x^9 - 81x^8 + 144x^7 + 182x^6 - 345x^5 - 217x^4 + 498x^3 + 44x^2 - 315x + 105.$$

and $\text{Res}(x, T_{14}(x), x) = 105 \mid T'_{14}(0) = -315$ and $\text{Res}(x+1, T_{14}(x), x) = 105 \mid T'_{14}(-1) = 315$ and $\text{Res}(2x+1, T_{14}(x), x) = 860055 = 3 \times 5 \times 7 \times 8191$ divides $2^{12} T'_{14}\left(-\frac{1}{2}\right) = 0$. $\square$

The values for D and d are listed in Table 2. We will also need the following lemmas.

k	D divides	d divides	C_k
2	1	1	$2 \cdot 3$
4	1	7	$2 \cdot 3 \cdot 5$
6	1	31	$2 \cdot 3 \cdot 7$
8	3	$3 \cdot 127$	$2 \cdot 3^2 \cdot 5$
10	5	$5 \cdot 7 \cdot 73$	$2 \cdot 3 \cdot 11$
12	691	$23 \cdot 89 \cdot 691$	$2 \cdot 3 \cdot 5 \cdot 7 \cdot 13$
14	$3 \cdot 5 \cdot 7$	$3 \cdot 5 \cdot 7 \cdot 8191$	$2 \cdot 3^2 \cdot 5$

TABLE 1. Summary of constants

Lemma 2.3. Suppose

$$ab = dy^n$$

for $a, b, d, y, n \in \mathbb{N}$ where $\gcd(a, b) = 1$. Then there exist $d_1, d_2, y_1, y_2 \in \mathbb{N}$ such that

$$\begin{aligned} d &= d_1 d_2, & y &= y_1 y_2 \\ \gcd(d_1, d_2) &= \gcd(y_1, y_2) = 1 \\ a/d_1 &= y_1^n, & b/d_2 &= y_2^n. \end{aligned}$$

Proof. First note that d is a divisor of ab . Since $\gcd(a, b) = 1$, the divisors of ab are in bijection with (d_1, d_2) , where d_1, d_2 are divisors of a, b , respectively. In this bijection, we have that $d_1 = \gcd(d, a)$, $d_2 = \gcd(d, b)$, and $d = d_1 d_2$. Now write

$$(5) \quad (a/d_1)(b/d_2) = y^n.$$

Since $\gcd(a/d_1, b/d_2) = 1$, we have that the valuation of each prime dividing the left hand side of (5) is divisible by n . Hence, by unique factorization in $\mathbb{Z}$, we obtain that $a/d_1 = y_1^n$ and $b/d_2 = y_2^n$ for some $y_1, y_2 \in \mathbb{N}$. $\square$

Proposition 2.4. For even k , suppose

$$S_k(x) = \frac{1}{C_k} x(x+1)(2x+1)T_k(x) = y^n$$

has solution x, y, n in $\mathbb{N}$. Let $D = \gcd(x(x+1), T_k(x))$ and $d = \gcd(2x+1, \frac{T_k(x)}{D})$. Assume

- (i) D and d are both square-free,
- (ii) if a prime $p \mid D$ or $p \mid d$, then $p \parallel T_k(x)$.

Then for some $\alpha, \beta, \alpha_0, \alpha_1, \beta_0, D_1, d_1 \in \mathbb{N}$ with $\alpha_0, \alpha_1, \beta_0, D, d$ pairwise coprime satisfying

$$\alpha\beta = \frac{C_k}{2 \gcd(D, C_k)}, \quad \alpha_0\alpha_1 = 2\alpha, \quad \beta_0 \mid \frac{\beta}{\gcd(d, \beta)},$$

the system of equations

$$(6) \quad \alpha_1 w_1 - \alpha_0 w_0 - 1 = 0$$

$$(7) \quad \beta_0 w_2 - 2\alpha_0 w_0 - 1 = 0$$

$$(8) \quad 2\alpha_1 w_1 - \beta_0 w_2 - 1 = 0$$

is solvable in $u_0, u_1, v_0 \in \mathbb{N}$ where

$$(9) \quad w_0 = \begin{cases} (D_1)^{n-1} u_0^n & \text{if } (D_1)^{n-1} \mid x \\ u_0^n & \text{if } (D_1)^{n-1} \nmid x \end{cases}, \quad w_1 = \begin{cases} u_1^n & \text{if } (D_1)^{n-1} \mid x \\ (D_1)^{n-1} u_1^n & \text{if } (D_1)^{n-1} \nmid x \end{cases}, \quad w_2 = d_1^{n-1} v_0^n.$$

and $D_1 = \frac{D}{\gcd(D, C_k)}$ and $d_1 = \frac{d}{\gcd(d, \beta)}$. Further, $x = \alpha_0 w_0$, $x+1 = \alpha_1 w_1$, and $2x+1 = \beta_0 w_2$.

Proof. Note that since $D \mid T_k(x)$, we have $D \mid C_k y^n$. Next, define $D_1 := \frac{D}{\gcd(D, C_k)}$ so that $\gcd(D_1, C_k) = 1$. Since $D_1 \mid D$, $D \mid C_k y^n$ and $\gcd(D_1, C_k) = 1$, we have $D_1 \mid y^n$. Additionally, since D is squarefree, so is D_1 , thus $D_1 \mid y$. Let $y_m := \frac{y}{D_1}$.

Multiplying (4) by $\frac{C_k}{D}$, we have

$$\begin{aligned} x(x+1)(2x+1) \frac{T_k(x)}{D} &= \frac{C_k y^n}{D} \\ &= \frac{C_k y^n}{\gcd(D, C_k) D_1} \\ &= \frac{C_k}{\gcd(D, C_k)} \frac{y^n}{D_1} \\ &= \frac{C_k}{\gcd(D, C_k)} (D_1)^{n-1} y_m^n. \end{aligned}$$

Since x , $x + 1$, and $2x + 1$ are pairwise coprime, and by Corollary 2.2,

$$(10) \quad \gcd\left(x(x+1), (2x+1)\frac{T_k(x)}{D}\right) = 1.$$

Next, we know $(D_1)^{n-1} \mid x(x+1)(2x+1)\frac{T_k(x)}{D}$. Since we have $D_1 \mid D \mid x(x+1)$, if $D_1 \mid (2x+1)\frac{T_k(x)}{D}$, then this would contradict (10). Hence $\gcd(D_1, (2x+1)\frac{T_k(x)}{D}) = 1$ so $(D_1)^{n-1} \mid x(x+1)$. Note that $2 \mid x(x+1)$. Then we have

$$(11) \quad \gcd\left(\frac{1}{2(D_1)^{n-1}}x(x+1), (2x+1)\frac{T_k(x)}{D}\right) = 1$$

so by Lemma 2.3 we can write

$$(12) \quad y_m^n = u^n v^n,$$

where

$$(13) \quad u^n = \frac{1}{2(D_1)^{n-1}\alpha}x(x+1), \quad v^n = \frac{1}{\beta}(2x+1)\frac{T_k(x)}{D}$$

where $\alpha\beta = \frac{C_k}{2\gcd(D, C_k)}$. Note that since $\gcd(x, x+1) = 1$, we must have either $\gcd(D_1, x) = 1$ or $\gcd(D_1, x+1) = 1$ and hence when $D_1 \neq 1$, we have (exclusively) $(D_1)^{n-1} \mid x$ or $(D_1)^{n-1} \mid x+1$. Again by Lemma 2.3, since $x, x+1$ are coprime, we can write

$$(14) \quad u^n = u_0^n u_1^n$$

where either

$$(a) \quad u_0^n = \frac{1}{\alpha_0(D_1)^{n-1}}x, \quad u_1^n = \frac{1}{\alpha_1}(x+1)$$

when $D_1 \mid x$ or

$$(b) \quad u_0^n = \frac{1}{\alpha_0}x, \quad u_1^n = \frac{1}{\alpha_1(D_1)^{n-1}}(x+1)$$

when $D_1 \mid x+1$, where $\alpha_0\alpha_1 = 2\alpha$. Define

$$w_0 = \begin{cases} (D_1)^{n-1}u_0^n & \text{if } D_1 \mid x \\ u_0^n & \text{if } D_1 \mid x+1 \end{cases}, \quad w_1 = \begin{cases} u_1^n & \text{if } D_1 \mid x \\ (D_1)^{n-1}u_1^n & \text{if } D_1 \mid x+1 \end{cases}$$

Rearranging and substituting the expressions for u_0^n and u_1^n , we get

$$(15) \quad \alpha_0 w_0 + 1 = \alpha_1 w_1.$$

It is clear that α_0, α_1, D_1 are coprime. Note that if $(D, d) \neq 1$, then there is a prime p such that $p \mid \gcd(x(x+1), 2x+1)$, which is a contradiction. Hence, $\gcd(d, D) = 1$.

Next, define $d_1 = \frac{d}{\gcd(d, \beta)}$. Since d is squarefree, so is d_1 . We have $\gcd(d_1, \beta) = 1$ and $d_1 \mid \beta v^n$ so $d_1 \mid v^n$. Since d_1 is squarefree, $d_1 \mid v$. Let $v_d := \frac{v}{d_1}$. Then

$$\begin{aligned} (2x+1) \frac{T_k(x)}{dD} &= \frac{\beta v^n}{d} \\ &= \frac{\beta}{\gcd(d, \beta)} \frac{v^n}{d_1} \\ (2x+1) \frac{T_k(x)}{dD} &= \frac{\beta}{\gcd(d, \beta)} d_1^{n-1} v_d^n \end{aligned}$$

By Corollary 2.2, we have

$$\gcd\left(2x+1, \frac{T_k(x)}{dD}\right) = 1.$$

Since $d_1 \mid d$, $\gcd\left(d_1, \frac{T_k(x)}{dD}\right) = 1$. Hence $d_1^{n-1} \mid 2x+1$ so

$$\gcd\left(\frac{1}{d_1^{n-1}}(2x+1), \frac{T_k(x)}{dD}\right) = 1$$

So by Lemma 2.3, we can write $v_d^n = v_0^n v_1^n$ where

$$(16) \quad v_0^n := \frac{1}{d_1^{n-1} \beta_0} (2x+1), \quad v_1^n := \frac{1}{\beta_1} \frac{T_k(x)}{dD}$$

and $\beta_0 \beta_1 = \frac{\beta}{\gcd(d, \beta)}$. Define

$$w_2 = d_1^{n-1} v_0^n$$

Rearranging and substituting w_0 and w_1 , we get

$$(17) \quad 2\alpha_0 w_0 + 1 = \beta_0 w_2$$

and

$$(18) \quad 2\alpha_1 w_1 - 1 = \beta_0 w_2$$

From each equation, we respectively see that $\alpha_0, D_1, \beta_0, d_1$ and $\alpha_1, D_1, \beta_0, d_1$ are pairwise coprime, proving the theorem. $\square$

3. THE MODULAR METHOD

Recall it suffices to consider $n = p$ prime. Using Proposition 2.4, solutions to (1) give rise to solutions to (6)–(8) which are of generalized Fermat type with one of the variables set to 1.

We now briefly summarize the modular method as it is applied to resolve generalized Fermat type equations of the form

$$(19) \quad Ax^p + By^p + Cz^p = 0,$$

where A, B, C are non-zero pairwise coprime integers which are not p th powers and such that $p \nmid ABC$. Our treatment follows [HK02].

Up to permutation of the terms and negation in (19), we can assume that By^p is even and $Ax^p \equiv -1 \pmod{4}$.

To a primitive non-trivial solution (a, b, c) of (19) we attach the Frey elliptic curve

$$(20) \quad Y^2 = X(X - Aa^p)(X + Bb^p)$$

Consider the Galois representation

$$\bar{\rho}_{E,p} : G_{\mathbb{Q}} \rightarrow \mathrm{GL}_2(\mathbb{F}_p)$$

attached to the elliptic curve E over $\mathbb{Q}$, which is obtained by letting $G_{\mathbb{Q}} := \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ act linearly on $E[p]$. As the E is a Frey elliptic curve over $\mathbb{Q}$, we have by modularity and level lowering that

$$(21) \quad \bar{\rho}_{E,p} \simeq \bar{\rho}_{f,\mathfrak{p}}$$

where f is a classical newform of weight 2, level N equal to the Serre level of $\bar{\rho}_{E,p}$, and trivial character and $\mathfrak{p}$ is a prime of the coefficient field K of f above p .

By Kraus-Halberstadt [HK02], we can compute the Serre level N of $\bar{\rho}_{E,p}$. Let $R = ABC$. Then

$$(22) \quad N = \begin{cases} 2\mathrm{rad}'(R) & v_2(R) = 0 \text{ or } v_2(R) \geq 5 \\ 2\mathrm{rad}'(R) & 1 \leq v_2(R) \leq 3, xyz \text{ even} \\ \mathrm{rad}'(R) & v_2(R_i) = 4 \\ 32\mathrm{rad}'(R) & v_2(R) = 1, xyz \text{ odd} \\ 8\mathrm{rad}'(R) & v_2(R) \in \{2, 3\}, xyz \text{ odd} \end{cases}.$$

The conductor N_E of $E/\mathbb{Q}$ is square-free except powers of 2. Hence, the Frey elliptic curve has either good or multiplicative bad reduction modulo primes $\ell \neq 2$.

Computing the finite set of newforms f of weight 2, level N , and trivial character we now consider the isomorphism in (21) for each possible f . The idea behind the modular method is to show this is not possible for each f , i.e. eliminate every newform f as a possibility in the isomorphism (21) for p sufficiently large.

Let $\ell \nmid 2Np$ be a prime so that $\bar{\rho}_{E,p}$ is unramified at ℓ . Let $\mathrm{Frob}_{\ell} \in G_{\mathbb{Q}} := \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ be a Frobenius element at ℓ . We may thus consider the well defined trace value

$$(23) \quad a_{\ell}(f) = \mathrm{tr} \rho_{f,\mathfrak{p}}(\mathrm{Frob}_{\ell}) \in \mathcal{O}_K,$$

where $\rho_{f,\mathfrak{p}}$ is the $\mathfrak{p}$ -adic representation attached to the newform f .

If $\ell \nmid N_E$, where N_E is the conductor of $E/\mathbb{Q}$, then E has good reduction at ℓ and we may consider

$$(24) \quad a_{\ell}(E) = \mathrm{tr} \rho_{E,p}(\mathrm{Frob}_{\ell}) \in \mathbb{Z},$$

where $\rho_{E,p}$ is the p -adic representation attached to E .

The isomorphism (21) implies that

$$(25) \quad a_{\ell}(E) \equiv a_{\ell}(f) \pmod{\mathfrak{p}},$$

and hence

$$(26) \quad \text{Norm}_{K/\mathbb{Q}}(a_\ell(E) - a_\ell(f)) \equiv 0 \pmod{p}$$

where $\text{Norm}_{K/\mathbb{Q}}(\cdot)$ is the norm from K to $\mathbb{Q}$. If the above norm is non-zero, we obtain a bound on the prime exponent p , i.e. for p sufficiently large, (21) does not hold for the newform f .

If $\ell \mid N_E$, then E has multiplicative bad reduction at ℓ . It can be shown that $\text{WD}_\ell(\{\rho_{E,p}\})$ is a special representation with character being an unramified character of order dividing 2. It follows that

$$(27) \quad \pm(\ell + 1) \equiv a_\ell(f) \pmod{\mathfrak{p}},$$

and hence

$$(28) \quad \text{Norm}_{K/\mathbb{Q}}((\ell + 1)^2 - a_\ell(f)^2) \equiv 0 \pmod{p}.$$

which gives a bound on the prime exponent p as the above norm is non-zero by the Deligne-Ramanujan bounds, i.e. for p sufficiently large, (21) does not hold for the newform f .

Proposition 3.1. Let (a, b, p) be a solution to

$$(29) \quad Ax^n - By^n = 1$$

where A, B, C are pairwise coprime. WLOG, we may assume $Bb^p \equiv 0 \pmod{2}$. Let $\mathcal{N}$ be the set of possible levels given by (22), ℓ a prime, and W_ℓ be the set of solutions to (29) mod ℓ . For $w = (w_0, w_1, n) \in W_\ell$, define $E(w) : y^2 = X(X+1)(X - Bw_1^n)$. Let $S_2(N)$ be the set of weight 2 newforms of level N and for $f \in S_2(N)$,

$$B_\ell(f, w) := \begin{cases} \text{Norm}_{K/\mathbb{Q}}(a_\ell(E(w)) - a_\ell(f)) & E(w) \text{ has good reduction at } \ell, \\ \text{Norm}_{K/\mathbb{Q}}((\ell + 1)^2 - a_\ell(f)^2) & E(w) \text{ has bad reduction at } \ell. \end{cases}$$

Then,

$$p \in \bigcup_{N \in \mathcal{N}} \bigcup_{f \in S_2(N)} \bigcap_{\substack{\ell \nmid N \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell(f, w)\} \right) \cup \{\ell\}$$

for any bound $L \in \mathbb{N}$.

Proof. If (a, b, p) is a solution to (29) then by the above discussion, there is a newform f of weight 2 with level $N \in \mathcal{N}$ such that $\bar{\rho}_{E, \ell} \simeq \bar{\rho}_{f, \ell}$, where $E : y^2 = X(X+1)(X - Bb^p)$. Since (a, b, p) is a solution to (29), $\exists w = (w_0, w_1, n) \in W_\ell$ where $w_0^n \equiv a^p \pmod{\ell}$ and $w_1^n \equiv b^p \pmod{\ell}$. By (26) and (28), we have thus have $p \mid B_\ell(f, w)$ or $p = \ell$. In particular,

$$p \in \{p \text{ prime} : p \mid B_\ell(f, w)\} \cup \{\ell\}$$

and thus

$$p \in \bigcup_{N \in \mathcal{N}} \bigcup_{f \in S_2(N)} \bigcap_{\substack{\ell \nmid N \\ \ell \leq L}} \bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell(f, w)\} \cup \{\ell\}$$

□

In particular, if for all possible levels $N \in \mathcal{N}$ and all weight 2 newforms f of level N , there exists $\ell \nmid N$ prime such that $\forall w \in W_\ell$, $\{p \text{ prime} : p \mid B_\ell(f, w)\}$ is finite, then we have that p is one of only finitely many possibilities.

Instead consider a solution (a, b, c, p) to the system (6)–(8). Reducing modulo a prime ℓ , we get a solution $w = (w_0, w_1, w_2, n)$ to the system mod ℓ . Then the Frey elliptic curve $E_i(w)$ associated to each generalized Fermat type equation (6)–(8) after reducing mod ℓ is given by:

$$(30) \quad E_0(w) : y^2 = \begin{cases} x(x+1)(x - \alpha_0 w_0^n) & \alpha_0 \equiv 0 \pmod{2} \\ x(x+1)(x + \alpha_1 w_1^n) & \alpha_1 \equiv 0 \pmod{2} \end{cases},$$

$$(31) \quad E_1(w) : y^2 = x(x+1)(x - 2\alpha_0 w_0^n),$$

$$(32) \quad E_2(w) : y^2 = x(x+1)(x + 2\alpha_1 w_1^n).$$

The quantities R_i are given by:

$$(33) \quad R_0 = \alpha_0 \alpha_1 D_1, \quad R_1 = \begin{cases} 2\alpha_0 D_1 \beta_0 d_1 & \text{if } D_1 \mid x \\ 2\alpha_0 \beta_0 d_1 & \text{if } D_1 \mid x+1 \end{cases}, \quad R_2 = \begin{cases} 2\alpha_1 \beta_0 d_1 & \text{if } D_1 \mid x \\ 2\alpha_1 D_1 \beta_0 d_1 & \text{if } D_1 \mid x+1 \end{cases}$$

from which we can compute the Serre levels N_i of $\bar{\rho}_{E_i, p}$ by (22).

For a newform f_i of level N_i , let $K_i = \mathbb{Q}(a_n(f_i) : n \in \mathbb{N})$. Recall from (21) and above, there exists a prime ideal $\mathfrak{p}_i$ of K_i over p such that for all primes $\ell \neq p$ and $\ell \nmid N_i$,

- (1) if E_i has good reduction at ℓ , then $a_\ell(E_i) \equiv a_\ell(f_i) \pmod{\mathfrak{p}_i}$,
- (2) if E_i has bad reduction at ℓ , then $\pm(\ell+1) \equiv a_\ell(f_i) \pmod{\mathfrak{p}_i}$.

For a solution $w = (w_0, w_1, w_2, n)$ to the system (6)–(8) modulo a prime ℓ , define

$$B_\ell^{(i)}(f, w) := \begin{cases} \text{Norm}_{K_i/\mathbb{Q}}(a_\ell(E_i) - a_\ell(f)) & E_i(w) \text{ good reduction at } \ell, \\ \text{Norm}_{K_i/\mathbb{Q}}((\ell+1)^2 - a_\ell(f))^2 & E_i(w) \text{ bad reduction at } \ell. \end{cases}$$

Thus any solution to (6)–(8) must have exponent $p \mid B_i(f)$ or $p = \ell$. Note that since this holds for all primes $\ell \neq p$, we can consider $B_i(f)$ for various ℓ and take the intersection of primes which appear in $B_i(f)$ to restrict the exponent p : we have a finite bound on the prime exponent p for the given f , provided $B_\ell^{(i)}(f) \neq 0$ for some $i = 0, 1, 2$.

Corollary 3.2. Let (x, y, p) be a non-trivial solution to (1). Let $\mathcal{N}_i$ be the set of possible levels to the elliptic curves in (30) given by Kraus, $\mathcal{N} := \mathcal{N}_0 \cup \mathcal{N}_1 \cup \mathcal{N}_2$, ℓ a prime, and W_ℓ be the set of solutions of (6)–(8) respectively given by Proposition 2.4 mod $\ell \nmid \prod_{N \in \mathcal{N}} N$. For $w = (w_0, w_1, w_2, n) \in W_\ell$, let

$$E_0(w) : y^2 = \begin{cases} x(x+1)(x - \alpha_0 w_0) & \alpha_0 \equiv 0 \pmod{2} \\ x(x+1)(x + \alpha_1 w_1) & \alpha_1 \equiv 0 \pmod{2} \end{cases},$$

$$E_1(w) : y^2 = x(x+1)(x - 2\alpha_0 w_0),$$

$$E_2(w) : y^2 = x(x+1)(x + 2\alpha_1 w_1).$$

For $N_i \in \mathcal{N}_i$, let $S_2(N_i)$ be the set of weight 2 newforms of level N_i and for $f_i \in S_2(N_i)$, define

$$B_\ell^{(i)}(f_i, w) := \begin{cases} \text{Norm}_{K_i/\mathbb{Q}}(a_\ell(E_i(w)) - a_\ell(f_i)) & E_i(w) \text{ has good reduction at } \ell, \\ \text{Norm}_{K_i/\mathbb{Q}}((\ell+1)^2 - a_\ell(f_i))^2 & E_i(w) \text{ has bad reduction at } \ell. \end{cases}$$

and let

$$A_\ell^{(i)}(w) = \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w) \text{ for some newform } f_i \text{ of level } N_i \in \mathcal{N}_i\}$$

for $i = 0, 1, 2$. Then for any $i = 0, 1, 2$,

$$(34) \quad p \in \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{f \in S_2(N_i)} \bigcap_{\substack{\ell \nmid N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f, w)\} \right) \cup \{\ell\}$$

Further,

$$(35) \quad p \in \bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{f_0 \in S_2(N_0)} \bigcap_{\substack{\ell \nmid \prod_{N \in \mathcal{N}} N \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_\ell^{(1)}(w) \cap A_\ell^{(2)}(w) \right) \cup \{\ell\}$$

In particular, if for every possible system of equations given by Proposition 2.4, we can show that any of the sets given by Corollary 3.2 are empty, then there are no solutions to (1).

Proof. For a non-trivial solution (x, y, p) to (1), by Proposition 2.4 we get a system

$$\begin{aligned} \alpha_1 w_1 - \alpha_0 w_0 &= 1 \\ \beta_0 w_2 - 2\alpha_0 w_0 &= 1 \\ 2\alpha_1 w_1 - \beta_0 w_2 &= 1 \end{aligned}$$

where

$$w_0 = \begin{cases} (D_1)^{n-1} u_0^n & \text{if } (D_1)^{n-1} \mid x \\ u_0^n & \text{if } (D_1)^{n-1} \nmid x \end{cases}, \quad w_1 = \begin{cases} u_1^n & \text{if } (D_1)^{n-1} \mid x \\ (D_1)^{n-1} u_1^n & \text{if } (D_1)^{n-1} \nmid x \end{cases}, \quad w_2 = d_1^{n-1} v_0^n.$$

where α_i, β_0, D, d are pairwise coprime. Applying Proposition 3.1 to (6), (7), and (8), we have that

$$p \in \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{f \in S_2(N_i)} \bigcap_{\substack{\ell \nmid N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell^{(i)}} \{p \text{ prime} : p \mid B_\ell^{(i)}(f, w)\} \right) \cup \{\ell\}$$

for $i = 0, 1, 2$ respectively, where $W_\ell^{(i)}$ are the solutions to (6), (7), and (8) mod ℓ . Since the solutions (x, y, p) must satisfy all three equations simultaneously, for a given prime ℓ , any solution w to the system mod ℓ must be in $W_\ell = W_\ell^{(0)} \cap W_\ell^{(1)} \cap W_\ell^{(2)}$. Thus,

$$p \in \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{f \in S_2(N_i)} \bigcap_{\substack{\ell \nmid N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f, w)\} \right) \cup \{\ell\}$$

Note for $i = 0, 1, 2$,

$$\begin{aligned}
& \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{f_i \in S_2(N_i)} \bigcap_{\substack{\ell \nmid N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\} \right) \cup \{\ell\} \\
& \subseteq \bigcap_{\substack{\ell \nmid \prod_{N_i \in \mathcal{N}_i} N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \bigcup_{N_i \in \mathcal{N}} \bigcup_{f_i \in S_2(N_i)} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\} \right) \cup \{\ell\} \\
& = \bigcap_{\substack{\ell \nmid \prod_{N_i \in \mathcal{N}_i} N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} A_\ell^{(i)}(w) \right) \cup \{\ell\}
\end{aligned}$$

Hence,

$$\begin{aligned}
p & \in \bigcap_{i=0}^2 \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{f_i \in S_2(N_i)} \bigcap_{\substack{\ell \nmid \prod_{N_i \in \mathcal{N}_i} N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\} \right) \cup \{\ell\} \\
& \subseteq \left(\bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{f_0 \in S_2(N_0)} \bigcap_{\substack{\ell \nmid \prod_{N_i \in \mathcal{N}_i} N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \right) \cup \{\ell\} \right) \cap \left(\bigcap_{i=0}^1 \bigcap_{\substack{\ell \nmid \prod_{N_i \in \mathcal{N}_i} N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} A_\ell^{(i)}(w) \right) \cup \{\ell\} \right) \\
& = \bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{f_0 \in S_2(N_0)} \bigcap_{\substack{\ell \nmid \prod_{N_i \in \mathcal{N}_i} N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_1(w) \cap A_2(w) \right) \cup \{\ell\}
\end{aligned}$$

□

Theorem 3.3. (*Darmon-Merel* [DM97]) The only non-zero primitive solutions (a, b, c) to the equation

$$(36) \quad x^n + y^n = 2z^n$$

satisfies $abc = \pm 1$.

Corollary 3.4. Suppose (x, y, n) is a solution to (1). By Proposition 2.4, there exist $\alpha_0, \alpha_1, \beta_0, D_1, d_1 \in \mathbb{N}$ pairwise coprime satisfying (6), (7), and (8). None of the following are true.

- (1) $\alpha_0 \alpha_1 = 2$ and $D_1 = 1$,
- (2) $\alpha_0 \beta_0 d_1 D_1 = 1$, or
- (3) $\alpha_0 \beta_0 d_1 D_1 = 1$

Proof. If $\alpha_0 \alpha_1 = 2$ and $D_1 = 1$, then (6) is

$$2u_0^n + 1 = u_1^n \quad \text{or} \quad u_0^n + 1 = 2u_1^n$$

In either case, by Theorem 3.3, the only solution satisfies $|u_0 u_1| = 1$. If $\alpha_0 = 1$, then $u_0 = u_1 = 1$ so $x = \alpha_0 u_0^n = 1$, which is the trivial solution. If $\alpha_1 = 1$, then $u_0 = u_1 = -1$ so $x = \alpha_0 u_0^n = -2 \notin \mathbb{N}$.

If $\alpha_0 \beta_0 d_1 D_1 = 1$, then (7) is

$$2u_0^n + 1 = v_0^n$$

and if $\alpha_1\beta_0d_1D_1 = 1$, then (8) is

$$2u_1^n - 1 = v_0^n$$

so we get $u_0 = -1$ and $u_1 = 1$ respectively, which as before leads to a contradiction. $\square$

4. LOCAL CONSTRAINTS

To resolve a specific prime exponent p for a generalized Fermat equation, i.e. show there are no non-trivial primitive solutions (a, b, c) for that specific prime exponent p , we try the following method of Kraus [Kra98].

Corollary 4.1. Let (x, y, p) be a solution to (1). Let $\ell \equiv 1 \pmod{p}$ be a prime. Let W_ℓ be the set of p -th power solutions (w_0, w_1, w_2, p) to the system given by Proposition 2.4 mod ℓ and $\mathcal{N}$ be the set of possible levels given the equations (6), (7), and (8). Then

$$(37) \quad p \in \bigcap_{i=0}^2 \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{f_i \in S_2(N_i)} \bigcap_{\substack{\ell \nmid N_i \\ \ell \leq L}} \bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\}$$

where $B_\ell^{(i)}(w)$ is given in Corollary 3.2.

This follows immediately from Proposition 3.1. Being a p th power mod ℓ , where $\ell \equiv 1 \pmod{p}$, is a very restrictive constraint. For example, if $\ell = 2p + 1$, then the only p th powers modulo ℓ are ± 1 .

Another local constraint we can impose is the following:

Lemma 4.2. If (x, y, p) is a solution to (1), then there exists a solution (w_0, w_1, w_2, p) to the system from Proposition 2.4 modulo p^2 .

In particular, if we can show that there are no solutions (w_0, w_1, w_2, p) modulo p^2 , then we can eliminate p from being a possible exponent.

Let S be a finite set of primes. Write

$$(38) \quad K = \bigcap_{i=0}^2 \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{f_i \in S_2(N_i)} \bigcap_{\substack{\ell \nmid N_i \\ \ell \leq L}} \bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\}$$

as in Corollary 4.1 and let W_{p^2} be the set of solutions to the system (6)–(8) modulo p^2 . Then define

$$F(S) = \{p \in S : p \in K \text{ and } W_{p^2} \neq \emptyset\}$$

Proposition 4.3. Let S be a set of primes. For $p \in S$, if $\exists x, y \in \mathbb{N}$ such that (x, y, p) is a solution to (1) then $p \in F(S)$.

Proof. For $p \in S$, if $\exists x, y \in \mathbb{N}$ such that (x, y, p) is a solution to (1), then by Corollary 4.1, $p \in K$ and by Lemma 4.2, $W_{p^2} \neq \emptyset$. Thus $p \in F(S)$. $\square$

In particular, if S is a set of candidate primes that solve (1) and for $p \in S$, if we can show that $p \notin F(S)$, then p cannot solve (1). Letting S be the set of primes from Corollary 3.2, then we get the following.

Corollary 4.4. Considering the same setup as Corollary 3.2, we have that

$$(39) \quad p \in \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{f \in S_2(N_i)} \bigcap_{\substack{\ell \nmid N_i \\ \ell \leq L}} F \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f, w)\} \right) \cup \{\ell\}$$

and

$$(40) \quad p \in \bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{f_0 \in S_2(N_0)} \bigcap_{\substack{\ell \nmid \prod_{N \in \mathcal{N}} N \\ \ell \leq L}} F \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_\ell^{(1)}(w) \cap A_\ell^{(2)}(w) \right) \cup \{\ell\}$$

Proof. If (x, y, p) is a solution to (1), then by Corollary 3.2, we have that

$$p \in \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{f \in S_2(N_i)} \bigcap_{\substack{\ell \nmid N_i \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f, w)\} \right) \cup \{\ell\}$$

and

$$p \in \bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{f_0 \in S_2(N_0)} \bigcap_{\substack{\ell \nmid \prod_{N \in \mathcal{N}} N \\ \ell \leq L}} \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_\ell^{(1)}(w) \cap A_\ell^{(2)}(w) \right) \cup \{\ell\}$$

Applying Proposition 4.3 to the inner most unions of each, we get

$$p \in \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{f \in S_2(N_i)} \bigcap_{\substack{\ell \nmid N_i \\ \ell \leq L}} F \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f, w)\} \right) \cup \{\ell\}$$

and

$$p \in \bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{f_0 \in S_2(N_0)} \bigcap_{\substack{\ell \nmid \prod_{N \in \mathcal{N}} N \\ \ell \leq L}} F \left(\bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_\ell^{(1)}(w) \cap A_\ell^{(2)}(w) \right) \cup \{\ell\}$$

□

5. LINEAR FORMS IN LOGARITHMS

We will make use of two powerful theorems to deal with any cases not handled by the modular and local methods.

Theorem 5.1. [Ben01] If $A, B, n \in \mathbb{Z}$, $AB \neq 0$, $n \geq 3$, then

$$|Ax^n - By^n| = 1$$

has at most one solution in positive integers (x, y) .

In particular, if $A = B + 1$, then the only solution is $(x, y) = (1, 1)$.

Theorem 5.2. [BBGP10] If $1 < B \leq 400$, then all integer solutions (x, y, n) of

$$|x^n - By^n| = 1$$

with $|xy| > 1$, $n \geq 3$ and with $(B, n) \notin \{(235, 23), (282, 23), (295, 29), (329, 23), (354, 29)\}$ are given by

$n = 3$, $(B, x, y) = (7, \pm(2, 1)), (9 \pm (2, 1)), (17, \pm(18, 7)), (19, \pm(8, 3)),$
 $(20 \pm (19, 7)), (26, \pm(3, 1)), (63, \pm(4, 1)), (91, \pm(9, 2)), (124, \pm(5, 1)),$
 $(126, \pm(5, 1)), (182, \pm(17, 3)), (215, \pm(6, 1)), (217, \pm(6, 1)),$
 $(254, \pm(19, 3)), (342, \pm(7, 1)), (344, \pm(7, 1))$
 $n = 4$, $(B, x, y) = (5, \pm 3, \pm 2), (15, \pm 2, \pm 1), (17, \pm 2, \pm 1), (39, \pm 5, \pm 2),$
 $(80, \pm 3, \pm 1), (150, \pm 7, \pm 2), (255, \pm 4, \pm 1).$
 $n = 5$, $(B, x, y) = (31, \pm(2, 1)), (242, \pm(3, 1)), (244, \pm(3, 1))$
 $n = 6$, $(B, x, y) = (63, \pm 2, \pm 1)$
 $n = 7$, $(B, x, y) = (127, \pm(2, 1)), (129, \pm(2, 1)),$
 $n = 8$, $(B, x, y) = (255, \pm 2, \pm 1)$

6. THE EVEN CASES $k = 2, 4, 6, 8, 14$ AND n PRIME

By Proposition 2.4, for a fixed even k , if we can show that for all possible combinations of $\alpha_0, \alpha_1, \beta_0, D_1, d_1$ have no solutions, then there are no solutions to (1). Fix $k \in \{4, 6, 8, 14\}$ and $\alpha_0, \alpha_1, \beta_0, d_1, D_1$. If this case satisfies Corollary 3.4, then we move on. Otherwise, we apply either Corollary 4.4 to the whole system, or Proposition 3.1 to just one of the equations when the possible levels to some of the equations are computationally too large, with the bound $L = 1000$. We find that for each case, the only possible exponent is $p = 2$, except for the cases listed in the following table.

k	n	Case
$k = 4$	$n \in \mathbb{N}$	$\alpha_0 = 2, \alpha_1 = 3, \beta_0 = 5, D = 1 = d$
$k = 8, 14$	$n \in \mathbb{N}$	$\alpha_0 = 3, \alpha_1 = 2, \beta_0 = 5, D = 1 = d$
		$\alpha_0 = 2, \alpha_1 = 9, \beta_0 = 5, D = 1 = d$
		$\alpha_0 = 9, \alpha_1 = 2, \beta_0 = 5, D = 1 = d$
$k = 14$	$n = 3, 5, 7$	$\alpha_0 = 1, \alpha_1 = 18, \beta_0 = 1, D = 7, d = 1$
		$\alpha_0 = 18, \alpha_1 = 1, \beta_0 = 1, D = 7, d = 1$
		$\alpha_0 = 1, \alpha_1 = 90, \beta_0 = 1, D = 7, d = 1$
		$\alpha_0 = 90, \alpha_1 = 1, \beta_0 = 1, D = 7, d = 1$
		$\alpha_0 = 1, \alpha_1 = 10, \beta_0 = 1, D = 21, d = 1$
		$\alpha_0 = 10, \alpha_1 = 1, \beta_0 = 1, D = 21, d = 1$
		$\alpha_0 = 1, \alpha_1 = 18, \beta_0 = 1, D = 35, d = 1$
		$\alpha_0 = 18, \alpha_1 = 1, \beta_0 = 1, D = 35, d = 1$
		$\alpha_0 = 1, \alpha_1 = 18, \beta_0 = 1, D = 7, d = 8191$
		$\alpha_0 = 18, \alpha_1 = 1, \beta_0 = 1, D = 7, d = 8191$
		$\alpha_0 = 9, \alpha_1 = 10, \beta_0 = 1, D = 7, d = 8191$
		$\alpha_0 = 10, \alpha_1 = 9, \beta_0 = 1, D = 7, d = 8191$
		$\alpha_0 = 1, \alpha_1 = 10, \beta_0 = 1, D = 21, d = 8191$
		$\alpha_0 = 10, \alpha_1 = 1, \beta_0 = 1, D = 21, d = 8191$
		$\alpha_0 = 1, \alpha_1 = 18, \beta_0 = 1, D = 35, d = 8191$
		$\alpha_0 = 18, \alpha_1 = 1, \beta_0 = 1, D = 35, d = 8191$

In the first four exceptional cases, we could not eliminate any primes using the bound $L = 1000$, so we instead appeal to Theorem 5.1.

In the first and third exceptional cases, we have $\alpha_0 = 2, \beta_0 = 5$, so (7) leads to

$$4u_0^n + 1 = 5v_0^n$$

By Theorem 5.1, the only solution is $(u_0, v_0) = (1, 1)$. Then $x = \alpha_0 u_0^n = 2$. Also, $S_4(2) = 1 + 2^4 = 17$, $S_8(2) = 1 + 2^8 = 257$, and $S_{14}(2) = 16385 = 5 \cdot 29 \cdot 113$, all of which are squarefree so we must have $n = 1$. But we assumed $n = p$ is prime, so there are no solutions in these cases. In the second and fourth exceptional cases, we have $\alpha_1 = 2, \beta_0 = 5, d_1 = 1$, so (8) leads to

$$4u_1^n - 1 = 5v_0^n$$

and by Bennett [Ben01], the only solution is $(u_0, v_0) = (-1, -1)$, so $x = \alpha_1 u_1^n - 1 < 0$, which is a contradiction.

In the remaining exceptional cases, all primes except for $p \in \{2, 3, 5, 7\}$ were eliminated. In the cases with $\beta_0 = 1 = d$, if $D_1 \mid x$, (8) leads to

$$2\alpha_1 u_1^n - v_0^n = 1$$

and if $D_1 \mid x + 1$, (7) leads to

$$v_0^n - 2\alpha_0 u_0^n = 1$$

In either case [BBGP10] shows that there are no solutions when $n = 3, 5, 7$ and $2\alpha_i \mid 90$.

For the remaining cases, (6), gives

$$\alpha_1 u_1^n - \alpha_0 D_1^{n-1} u_0^n = 1$$

when $D_1 \mid x$ and

$$\alpha_1 D_1^{n-1} u_1^n - \alpha_0 u_0^n = 1$$

when $D_1 \mid x + 1$. For $n = 3, 5, 7$, MAGMA's Thue solver shows that there are no solutions to either equation.

Finally, Jacobson, Pintér, and Walsh [JPW03] have verified Schäffer's conjecture when $k \leq 58$ is even and $n = 2$. Thus, since we have shown that there are no solutions to the system (14), (15), (16), by Proposition 2.4 there are no solutions to $S_k(x) = y^n$ for $k = 4, 6, 8, 14$.

7. CONCLUSIONS

We have shown the cases $2 \leq k \leq 14$ even, except $k = 12$, of Schäffer's conjecture by systematically applying a more detailed descent, the modular method plus local methods, and then using prior results on ternary equations with coefficients.

The case $k = 12$ remains outside the reach of this approach as the levels in the modular method become too big.

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DEPARTMENT OF MATHEMATICS, SIMON FRASER UNIVERSITY, BURNABY, BC V5A 1S6, CANADA.

Email address: `ichen@sfu.ca`

DEPARTMENT OF MATHEMATICS, SIMON FRASER UNIVERSITY, BURNABY, BC V5A 1S6, CANADA.

Email address: `schoia@sfu.ca@sfu.ca`

DEPARTMENT OF MATHEMATICS, SIMON FRASER UNIVERSITY, BURNABY, BC V5A 1S6, CANADA.

Email address: `james_houle@sfu.ca`